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**Sent:** Tuesday, May 10, 2022 11:53 AM

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**Subject:** STN 125742/45: Revised PI labeling

(b) (6) Digitally signed by (b) (6)  
Date: 2022.05.10  
14:29:29 -04'00'

Kate,

We have reviewed the Package Insert that was submitted to STN 125742.45 on April 29<sup>th</sup> and we have a couple of additional proposed revisions in the attached document. Also, we discussed Pfizer's proposal to maintain the originally used chemical name and we are in agreement at this time. The review team has requested that the PI be sent back by Monday, May 16<sup>th</sup>. In addition, please also submit a Tris/Sucrose version of the PI (tracked changes and clean running).

Regards,

(b) (6)

- Please confirm receipt of this email and let us know if you have any questions.

(b) (6)

(b) (6)

**Food and Drug Administration  
Center for Biologics Evaluation & Research  
Office of Vaccines Research & Review**

(b) (6)



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## HIGHLIGHTS OF PRESCRIBING INFORMATION

These highlights do not include all the information needed to use COMIRNATY safely and effectively. See full prescribing information for COMIRNATY.

**COMIRNATY® (COVID-19 Vaccine, mRNA) suspension for injection, for intramuscular use**

Initial U.S. Approval: 2021

### RECENT MAJOR CHANGES

Indications and Usage (1) M/YYYY  
Dosage and Administration, Preparation for Administration (2.1) 12/2021

### INDICATIONS AND USAGE

COMIRNATY is a vaccine indicated for active immunization to prevent coronavirus disease 2019 (COVID-19) caused by severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) in individuals 12 years of age and older. (1)

### DOSAGE AND ADMINISTRATION

- COMIRNATY supplied in multiple dose vials with purple caps and labels with purple borders MUST BE DILUTED before use. (2.1)
- For intramuscular injection only. (2.2)
- COMIRNATY is administered intramuscularly as a series of 2 doses (0.3 mL each) 3 weeks apart. (2.3)

### DOSAGE FORMS AND STRENGTHS

Suspension for injection. After preparation, a single dose is 0.3 mL. (3)

### CONTRAINDICATIONS

Known history of a severe allergic reaction (e.g., anaphylaxis) to any component of COMIRNATY. (4)

### WARNINGS AND PRECAUTIONS

- Postmarketing data demonstrate increased risks of myocarditis and pericarditis, particularly within 7 days following the second dose. (5.2)
- Syncope (fainting) may occur in association with administration of injectable vaccines, including COMIRNATY. Procedures should be in place to avoid injury from fainting. (5.4)

### ADVERSE REACTIONS

- In clinical studies of participants 16 through 55 years of age, the most commonly reported adverse reactions ( $\geq 10\%$ ) were pain at the injection site (88.6%), fatigue (70.1%), headache (64.9%), muscle pain (45.5%), chills (41.5%), joint pain (27.5%), fever (17.8%), and injection site swelling (10.6%). (6.1)
- In clinical studies of participants 56 years of age and older, the most commonly reported adverse reactions ( $\geq 10\%$ ) were pain at the injection site (78.2%), fatigue (56.9%), headache (45.9%), muscle pain (32.5%), chills (24.8%), joint pain (21.5%), injection site swelling (11.8%), fever (11.5%), and injection site redness (10.4%). (6.1)
- In clinical studies of adolescents 12 through 15 years of age, the most commonly reported adverse reactions ( $\geq 8\%$ ) were pain at the injection site (90.5%), fatigue (77.5%), headache (75.5%), chills (49.2%), muscle pain (42.2%), fever (24.3%), joint pain (20.2%), injection site swelling (9.2%), and injection site redness (8.6%). (6.1)

To report SUSPECTED ADVERSE REACTIONS, contact Pfizer Inc. at 1-800-438-1985 or VAERS at 1-800-822-7967 or <http://vaers.hhs.gov>.

See 17 for PATIENT COUNSELING INFORMATION.

Revised: M/YYYY

## FULL PRESCRIBING INFORMATION: CONTENTS\*

### 1 INDICATIONS AND USAGE

### 2 DOSAGE AND ADMINISTRATION

- 2.1 Preparation for Administration
- 2.2 Administration Information
- 2.3 Vaccination Schedule

### 3 DOSAGE FORMS AND STRENGTHS

### 4 CONTRAINDICATIONS

### 5 WARNINGS AND PRECAUTIONS

- 5.1 Management of Acute Allergic Reactions
- 5.2 Myocarditis and Pericarditis
- 5.3 Syncope
- 5.4 Altered Immunocompetence
- 5.5 Limitation of Effectiveness

### 6 ADVERSE REACTIONS

- 6.1 Clinical Trials Experience
- 6.2 Postmarketing Experience

### 8 USE IN SPECIFIC POPULATIONS

- 8.1 Pregnancy

- 8.2 Lactation

- 8.4 Pediatric Use

- 8.5 Geriatric Use

### 11 DESCRIPTION

### 12 CLINICAL PHARMACOLOGY

- 12.1 Mechanism of Action

### 13 NONCLINICAL TOXICOLOGY

- 13.1 Carcinogenesis, Mutagenesis, Impairment of Fertility

### 14 CLINICAL STUDIES

- 14.1 Efficacy in Participants 16 Years of Age and Older
- 14.2 Efficacy in Adolescents 12 Through 15 Years of Age
- 14.3 Immunogenicity in Adolescents 12 Through 15 Years of Age

### 16 HOW SUPPLIED/STORAGE AND HANDLING

### 17 PATIENT COUNSELING INFORMATION

\* Sections or subsections omitted from the full prescribing information are not listed.

## FULL PRESCRIBING INFORMATION

### 1 INDICATIONS AND USAGE

COMIRNATY is a vaccine indicated for active immunization to prevent coronavirus disease 2019 (COVID-19) caused by severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) in individuals 12 years of age and older.

### 2 DOSAGE AND ADMINISTRATION

For intramuscular injection only.

#### 2.1 Preparation for Administration

The storage, preparation, and administration information in this Prescribing Information apply to COMIRNATY for individuals 16 years of age and older supplied in multiple dose vials with purple caps and labels with purple borders, which **MUST BE DILUTED** before use.

##### COMIRNATY Multiple Dose Vial with a Purple Cap and Label with a Purple Border

| Age Range          | Dilution Information                                                        | Doses Per Vial After Dilution | Dose Volume |
|--------------------|-----------------------------------------------------------------------------|-------------------------------|-------------|
| 16 years and older | Dilute with 1.8 mL sterile 0.9% Sodium Chloride Injection, USP prior to use | 6                             | 0.3 mL      |

##### Dose Preparation

Each vial **MUST BE DILUTED** before administering the vaccine.

##### Prior to Dilution

- COMIRNATY multiple dose vial with a purple cap and label with a purple border contains a volume of 0.45 mL, supplied as a frozen suspension that does not contain preservative.
- Each vial must be thawed before dilution.
- Vials may be thawed in the refrigerator [2°C to 8°C (35°F to 46°F)] or at room temperature [up to 25°C (77°F)] [*see How Supplied/Storage and Handling (16)*].
- Refer to thawing instructions in the panels below.

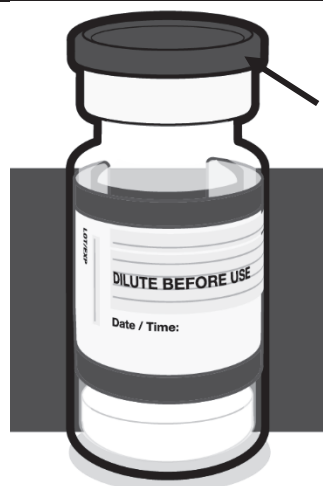
##### Dilution

- Dilute the vial contents using 1.8 mL of sterile 0.9% Sodium Chloride Injection, USP to form COMIRNATY. Do not add more than 1.8 mL of diluent.
- ONLY use sterile 0.9% Sodium Chloride Injection, USP as the diluent. Do not use bacteriostatic 0.9% Sodium Chloride Injection or any other diluent.
- Vials of sterile 0.9% Sodium Chloride Injection, USP are provided but shipped separately. Use the provided diluent or another sterile 0.9% Sodium Chloride Injection, USP as the diluent.
  - Provided diluent vials are single-use only; discard after 1.8 mL is withdrawn.
  - If another sterile 0.9% Sodium Chloride Injection, USP is used as the diluent, discard after 1.8 mL is withdrawn.

- Do not dilute more than 1 vial of COMIRNATY using the same diluent vial.
- After dilution, 1 vial of COMIRNATY contains 6 doses of 0.3 mL each.
- Refer to dilution and dose preparation instructions in the panels below.

### Dilution and Preparation Instructions

#### COMIRNATY Vial with Purple Cap and Label with Purple Border – Vial Verification



Purple cap

✓ Purple plastic cap and label with purple border.

- Verify that the vial of COMIRNATY has a purple plastic cap and a label with a purple border.

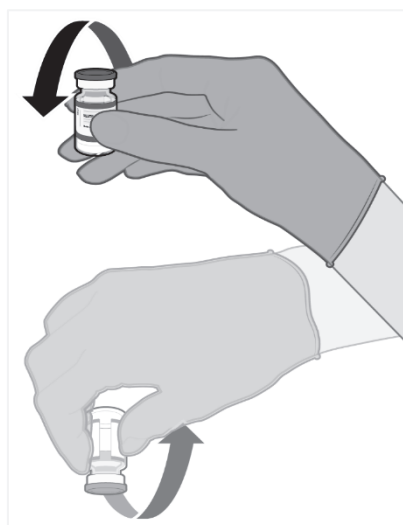
#### COMIRNATY Vial with Purple Cap and Label with Purple Border – Thawing Prior to Dilution



No more than 2 hours at room temperature (up to 25°C/77°F).

- Thaw vial(s) of COMIRNATY before dilution either by:
  - Allowing vial(s) to thaw in the refrigerator [2°C to 8°C (35°F to 46°F)]. A carton of vials may take up to 3 hours to thaw, and thawed vials can be stored in the refrigerator for up to 1 month.
  - Allowing vial(s) to sit at room temperature [up to 25°C (77°F)] for 30 minutes.
- Using either thawing method, vials must reach room temperature before dilution and must be diluted within 2 hours.

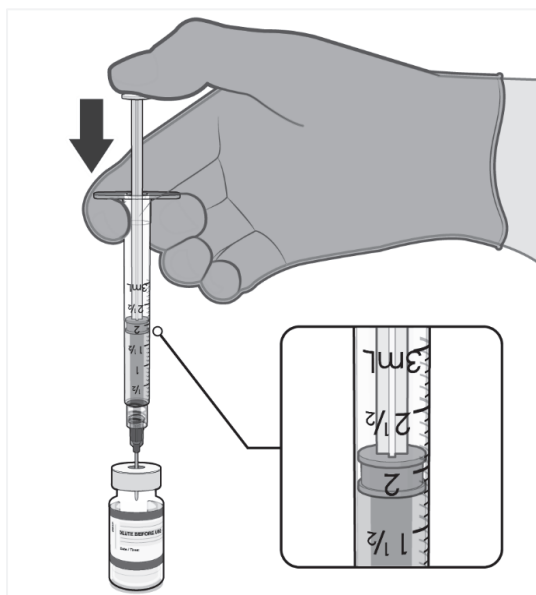
## Dilution and Preparation Instructions



Gently × 10

- Before dilution invert vaccine vial gently 10 times.
- Do not shake.
- Inspect the liquid in the vaccine vial prior to dilution. The liquid is a white to off-white suspension and may contain white to off-white opaque amorphous particles.
- Do not use if liquid is discolored or if other particles are observed.

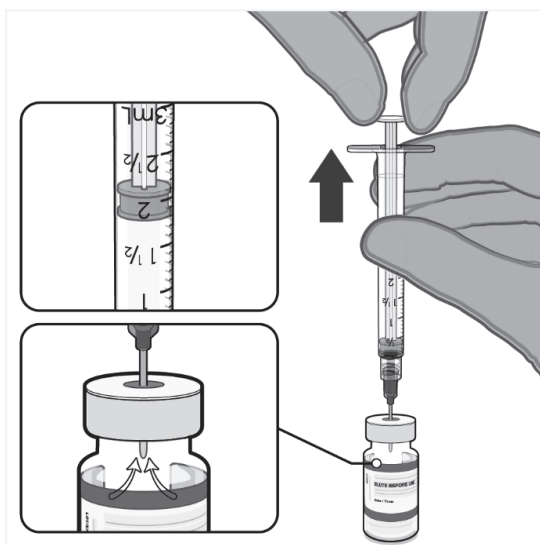
## COMIRNATY Vial with Purple Cap and Label with Purple Border – Dilution



Add 1.8 mL of sterile 0.9% sodium chloride injection, USP.

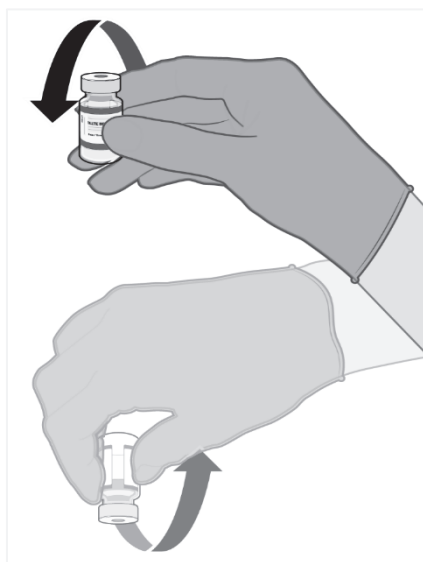
- ONLY use sterile 0.9% Sodium Chloride Injection, USP as the diluent.
- Withdraw 1.8 mL of diluent into a transfer syringe (21-gauge or narrower needle).
- Add 1.8 mL of sterile 0.9% Sodium Chloride Injection, USP into the vaccine vial.

## Dilution and Preparation Instructions



**Pull back plunger to 1.8 mL to remove air from vial.**

- Equalize vial pressure before removing the needle from the vaccine vial by withdrawing 1.8 mL air into the empty diluent syringe.



**Gently × 10**

- Gently invert the vial containing COMIRNATY 10 times to mix.
- Do not shake.
- Inspect the vaccine in the vial.
- The vaccine will be an off-white suspension. Do not use if vaccine is discolored or contains particulate matter.

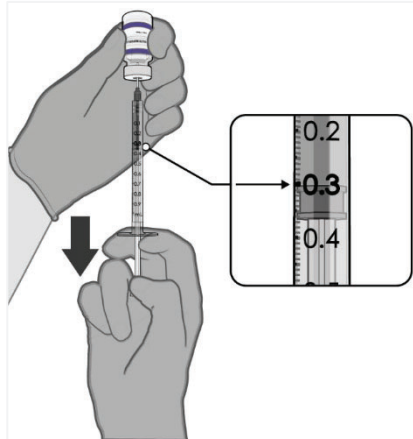
## Dilution and Preparation Instructions



**Record the date and time of dilution.  
Use within 6 hours after dilution.**

- Record the date and time of dilution on the COMIRNATY vial label.
- Store between 2°C to 25°C (35°F to 77°F).
- Discard any unused vaccine 6 hours after dilution.

## COMIRNATY Vial with Purple Cap and Label with Purple Border – Preparation of Individual 0.3 mL Doses



**Withdraw 0.3 mL dose of vaccine.**

- Withdraw 0.3 mL of COMIRNATY preferentially using low dead-volume syringes and/or needles.
- Each dose must contain 0.3 mL of vaccine.
- If the amount of vaccine remaining in a single vial cannot provide a full dose of 0.3 mL, discard the vial and any excess volume.
- Administer immediately.

## 2.2 Administration Information

Parenteral drug products should be inspected visually for particulate matter and discoloration prior to administration, whenever solution and container permit. The vaccine will be an off-white suspension. Do not administer if vaccine is discolored or contains particulate matter.

Administer a single 0.3 mL dose of COMIRNATY intramuscularly.

After dilution, vials of COMIRNATY with purple caps and labels with purple borders contain 6 doses of 0.3 mL of vaccine. Low dead-volume syringes and/or needles can be used to extract 6 doses from a single vial. If standard syringes and needles are used, there may not be sufficient volume to extract 6 doses from a single vial. Irrespective of the type of syringe and needle,

- each dose must contain 0.3 mL of vaccine.
- if the amount of vaccine remaining in the vial cannot provide a full dose of 0.3 mL, discard the vial and any excess volume.



- do not pool excess vaccine from multiple vials.

## **2.3 Vaccination Schedule**

COMIRNATY is administered intramuscularly as a series of 2 doses (0.3 mL each) 3 weeks apart.

There are no data available on the interchangeability of COMIRNATY with COVID-19 vaccines from other manufacturers to complete the vaccination series. Individuals who have received 1 dose of COMIRNATY should receive a second dose of COMIRNATY to complete the vaccination series.

## **3 DOSAGE FORMS AND STRENGTHS**

COMIRNATY is a suspension for injection. After preparation, each dose of COMIRNATY supplied in vials with purple caps and labels with purple borders is 0.3 mL.

## **4 CONTRAINDICATIONS**

Do not administer COMIRNATY to individuals with known history of a severe allergic reaction (e.g., anaphylaxis) to any component of the COMIRNATY [*see Description (11)*].

## **5 WARNINGS AND PRECAUTIONS**

### **5.1 Management of Acute Allergic Reactions**

Appropriate medical treatment used to manage immediate allergic reactions must be immediately available in the event an acute anaphylactic reaction occurs following administration of COMIRNATY.

### **5.2 Myocarditis and Pericarditis**

Postmarketing data demonstrate increased risks of myocarditis and pericarditis, particularly within 7 days following the second dose. The observed risk is higher among males under 40 years of age than among females and older males. The observed risk is highest in males 12 through 17 years of age. Although some cases required intensive care support, available data from short-term follow-up suggest that most individuals have had resolution of symptoms with conservative management. Information is not yet available about potential long-term sequelae. The CDC has published considerations related to myocarditis and pericarditis after vaccination, including for vaccination of individuals with a history of myocarditis or pericarditis (<https://www.cdc.gov/vaccines/covid-19/clinical-considerations/myocarditis.html>).

### **5.3 Syncope**

Syncope (fainting) may occur in association with administration of injectable vaccines, including COMIRNATY. Procedures should be in place to avoid injury from fainting.

### **5.4 Altered Immunocompetence**

Immunocompromised persons, including individuals receiving immunosuppressant therapy, may have a diminished immune response to the COMIRNATY.

## 5.5 Limitation of Effectiveness

COMIRNATY may not protect all vaccine recipients.

## 6 ADVERSE REACTIONS

In clinical studies, the most commonly reported ( $\geq 10\%$ ) adverse reactions in participants 16 through 55 years of age following any dose were pain at the injection site (88.6%), fatigue (70.1%), headache (64.9%), muscle pain (45.5%), chills (41.5%), joint pain (27.5%), fever (17.8%), and injection site swelling (10.6%).

In clinical studies, the most commonly reported ( $\geq 10\%$ ) adverse reactions in participants 56 years of age and older following any dose were pain at the injection site (78.2%), fatigue (56.9%), headache, (45.9%), muscle pain (32.5%), chills (24.8%), joint pain (21.5%), injection site swelling (11.8%), fever (11.5%), and injection site redness (10.4%).

In a clinical study, the most commonly reported ( $\geq 8\%$ ) adverse reactions in adolescents 12 through 15 years of age following any dose were pain at the injection site (90.5%), fatigue (77.5%), headache (75.5%), chills (49.2%), muscle pain (42.2%), fever (24.3%), joint pain (20.2%), injection site swelling (9.2%), and injection site redness (8.6%).

### 6.1 Clinical Trials Experience

Because clinical trials are conducted under widely varying conditions, adverse reaction rates observed in the clinical trials of a vaccine cannot be directly compared to rates in the clinical trials of another vaccine and may not reflect the rates observed in practice.

The safety of COMIRNATY was evaluated in participants 12 years of age and older in 2 clinical studies conducted in Germany (Study 1), United States, Argentina, Brazil, Turkey, South Africa, and Germany (Study 2). Study BNT162-01 (Study 1) was a Phase 1/2, 2-part, dose-escalation trial that enrolled 60 participants, 18 through 55 years of age and 36 participants, 56 through 85 years of age. Study C4591001 (Study 2) is a Phase 1/2/3 multicenter, multinational, randomized, saline placebo-controlled, double-blinded (Phase 2/3), dose-finding, vaccine candidate-selection and efficacy study that has enrolled approximately 46,000 participants 12 years of age or older. Of these, approximately 44,047 participants (22,026 COMIRNATY; 22,021 placebo) in Phase 2/3 are 16 years of age or older (including 378 and 376 participants 16 through 17 years of age in the COMIRNATY and placebo groups, respectively) and 2,260 adolescents are 12 through 15 years of age (1,131 and 1,129 in the COMIRNATY and placebo groups, respectively). Upon issuance of the Emergency Use Authorization for COMIRNATY, participants were unblinded to offer placebo participants COMIRNATY. Participants were unblinded in a phased manner over a period of months to offer placebo participants COMIRNATY. Study 2 also included 200 participants with confirmed stable human immunodeficiency virus (HIV) infection; HIV-positive participants are included in safety population disposition but are summarized separately in safety analyses. Confirmed stable HIV infection was defined as documented viral load  $<50$  copies/mL and CD4 count  $>200$  cells/mm<sup>3</sup> within 6 months before enrollment, and on stable antiretroviral therapy for at least 6 months.

In Study 2, all participants 12 through 15 years of age, and 16 years and older in the reactogenicity subset were monitored for solicited local and systemic reactions and use of antipyretic medication after each vaccination in an electronic diary. Participants are being monitored for unsolicited adverse events, including serious adverse events, throughout the study [from Dose 1 through 1 month (all unsolicited adverse events) or 6 months (serious adverse events) after the last vaccination]. Tables 1 through 6 present the frequency and severity of solicited local and systemic reactions, respectively, within 7 days following each dose of COMIRNATY and placebo.

## Participants 16 Years of Age and Older

At the time of the analysis of the ongoing Study 2 with a data cutoff of March 13, 2021, there were 25,651 (58.2%) participants (13,031 COMIRNATY and 12,620 placebo) 16 years of age and older followed for  $\geq 4$  months after the second dose.

Demographic characteristics in Study 2 were generally similar with regard to age, gender, race, and ethnicity among participants who received COMIRNATY and those who received placebo. Overall, among the total participants who received either COMIRNATY or placebo, 50.9% were male, 49.1% were female, 79.3% were 16 through 64 years of age, 20.7% were 65 years of age and older, 82.0% were White, 9.6% were Black or African American, 25.9% were Hispanic/Latino, 4.3% were Asian, and 1.0% were American Indian or Alaska Native.

### *Local and Systemic Adverse Reactions Solicited in the Study 2*

In participants 16 through 55 years of age after receiving Dose 2, the mean duration of pain at the injection site was 2.5 days (range 1 to 70 days), for redness 2.2 days (range 1 to 9 days), and for swelling 2.1 days (range 1 to 8 days) for participants in the COMIRNATY group. In participants 56 years of age and older after receiving Dose 2, the mean duration of pain at the injection site was 2.4 days (range 1 to 36 days), for redness 3.0 days (range 1 to 34 days), and for swelling 2.6 days (range 1 to 34 days) for participants in the COMIRNATY group.

**Table 1: Study 2 – Frequency and Percentages of Participants with Solicited Local Reactions, by Maximum Severity, Within 7 Days After Each Dose – Participants 16 Through 55 Years of Age – Reactogenicity Subset of the Safety Population\***

|                             | <b>COMIRNATY<br/>Dose 1<br/>N<sup>a</sup>=2899<br/>n<sup>b</sup> (%)</b> | <b>Placebo<br/>Dose 1<br/>N<sup>a</sup>=2908<br/>n<sup>b</sup> (%)</b> | <b>COMIRNATY<br/>Dose 2<br/>N<sup>a</sup>=2682<br/>n<sup>b</sup> (%)</b> | <b>Placebo<br/>Dose 2<br/>N<sup>a</sup>=2684<br/>n<sup>b</sup> (%)</b> |
|-----------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------|
| <b>Redness<sup>c</sup></b>  |                                                                          |                                                                        |                                                                          |                                                                        |
| Any (>2.0 cm)               | 156 (5.4)                                                                | 28 (1.0)                                                               | 151 (5.6)                                                                | 18 (0.7)                                                               |
| Mild                        | 113 (3.9)                                                                | 19 (0.7)                                                               | 90 (3.4)                                                                 | 12 (0.4)                                                               |
| Moderate                    | 36 (1.2)                                                                 | 6 (0.2)                                                                | 50 (1.9)                                                                 | 6 (0.2)                                                                |
| Severe                      | 7 (0.2)                                                                  | 3 (0.1)                                                                | 11 (0.4)                                                                 | 0                                                                      |
| <b>Swelling<sup>c</sup></b> |                                                                          |                                                                        |                                                                          |                                                                        |
| Any (>2.0 cm)               | 184 (6.3)                                                                | 16 (0.6)                                                               | 183 (6.8)                                                                | 5 (0.2)                                                                |
| Mild                        | 124 (4.3)                                                                | 6 (0.2)                                                                | 110 (4.1)                                                                | 3 (0.1)                                                                |
| Moderate                    | 54 (1.9)                                                                 | 8 (0.3)                                                                | 66 (2.5)                                                                 | 2 (0.1)                                                                |
| Severe                      | 6 (0.2)                                                                  | 2 (0.1)                                                                | 7 (0.3)                                                                  | 0                                                                      |

|                                               | <b>COMIRNATY<br/>Dose 1<br/>N<sup>a</sup>=2899<br/>n<sup>b</sup> (%)</b> | <b>Placebo<br/>Dose 1<br/>N<sup>a</sup>=2908<br/>n<sup>b</sup> (%)</b> | <b>COMIRNATY<br/>Dose 2<br/>N<sup>a</sup>=2682<br/>n<sup>b</sup> (%)</b> | <b>Placebo<br/>Dose 2<br/>N<sup>a</sup>=2684<br/>n<sup>b</sup> (%)</b> |
|-----------------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------|
| <b>Pain at the injection site<sup>d</sup></b> |                                                                          |                                                                        |                                                                          |                                                                        |
| Any                                           | 2426 (83.7)                                                              | 414 (14.2)                                                             | 2101 (78.3)                                                              | 312 (11.6)                                                             |
| Mild                                          | 1464 (50.5)                                                              | 391 (13.4)                                                             | 1274 (47.5)                                                              | 284 (10.6)                                                             |
| Moderate                                      | 923 (31.8)                                                               | 20 (0.7)                                                               | 788 (29.4)                                                               | 28 (1.0)                                                               |
| Severe                                        | 39 (1.3)                                                                 | 3 (0.1)                                                                | 39 (1.5)                                                                 | 0                                                                      |

Notes: Reactions were collected in the electronic diary (e-diary) from Day 1 to Day 7 after vaccination.

No Grade 4 solicited local reactions were reported in participants 16 through 55 years of age.

\* Randomized participants in the safety analysis population who received at least 1 dose of the study intervention. Participants with chronic, stable HIV infection were excluded.

a. N = Number of participants reporting at least 1 yes or no response for the specified reaction after the specified dose. The N for each reaction was the same, therefore, this information was included in the column header.

b. n = Number of participants with the specified reaction.

c. Mild: >2.0 to ≤5.0 cm; Moderate: >5.0 to ≤10.0 cm; Severe: >10.0 cm.

d. Mild: does not interfere with activity; Moderate: interferes with activity; Severe: prevents daily activity.

**Table 2: Study 2 – Frequency and Percentages of Participants with Solicited Systemic Reactions, by Maximum Severity, Within 7 Days After Each Dose – Participants 16 Through 55 Years of Age – Reactogenicity Subset of the Safety Population\***

|                             | <b>COMIRNATY<br/>Dose 1<br/>N<sup>a</sup>=2899<br/>n<sup>b</sup> (%)</b> | <b>Placebo<br/>Dose 1<br/>N<sup>a</sup>=2908<br/>n<sup>b</sup> (%)</b> | <b>COMIRNATY<br/>Dose 2<br/>N<sup>a</sup>=2682<br/>n<sup>b</sup> (%)</b> | <b>Placebo<br/>Dose 2<br/>N<sup>a</sup>=2684<br/>n<sup>b</sup> (%)</b> |
|-----------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------|
| <b>Fever</b>                |                                                                          |                                                                        |                                                                          |                                                                        |
| ≥38.0°C                     | 119 (4.1)                                                                | 25 (0.9)                                                               | 440 (16.4)                                                               | 11 (0.4)                                                               |
| ≥38.0°C to 38.4°C           | 86 (3.0)                                                                 | 16 (0.6)                                                               | 254 (9.5)                                                                | 5 (0.2)                                                                |
| >38.4°C to 38.9°C           | 25 (0.9)                                                                 | 5 (0.2)                                                                | 146 (5.4)                                                                | 4 (0.1)                                                                |
| >38.9°C to 40.0°C           | 8 (0.3)                                                                  | 4 (0.1)                                                                | 39 (1.5)                                                                 | 2 (0.1)                                                                |
| >40.0°C                     | 0                                                                        | 0                                                                      | 1 (0.0)                                                                  | 0                                                                      |
| <b>Fatigue<sup>c</sup></b>  |                                                                          |                                                                        |                                                                          |                                                                        |
| Any                         | 1431 (49.4)                                                              | 960 (33.0)                                                             | 1649 (61.5)                                                              | 614 (22.9)                                                             |
| Mild                        | 760 (26.2)                                                               | 570 (19.6)                                                             | 558 (20.8)                                                               | 317 (11.8)                                                             |
| Moderate                    | 630 (21.7)                                                               | 372 (12.8)                                                             | 949 (35.4)                                                               | 283 (10.5)                                                             |
| Severe                      | 41 (1.4)                                                                 | 18 (0.6)                                                               | 142 (5.3)                                                                | 14 (0.5)                                                               |
| <b>Headache<sup>c</sup></b> |                                                                          |                                                                        |                                                                          |                                                                        |
| Any                         | 1262 (43.5)                                                              | 975 (33.5)                                                             | 1448 (54.0)                                                              | 652 (24.3)                                                             |
| Mild                        | 785 (27.1)                                                               | 633 (21.8)                                                             | 699 (26.1)                                                               | 404 (15.1)                                                             |
| Moderate                    | 444 (15.3)                                                               | 318 (10.9)                                                             | 658 (24.5)                                                               | 230 (8.6)                                                              |
| Severe                      | 33 (1.1)                                                                 | 24 (0.8)                                                               | 91 (3.4)                                                                 | 18 (0.7)                                                               |
| <b>Chills<sup>c</sup></b>   |                                                                          |                                                                        |                                                                          |                                                                        |
| Any                         | 479 (16.5)                                                               | 199 (6.8)                                                              | 1015 (37.8)                                                              | 114 (4.2)                                                              |
| Mild                        | 338 (11.7)                                                               | 148 (5.1)                                                              | 477 (17.8)                                                               | 89 (3.3)                                                               |
| Moderate                    | 126 (4.3)                                                                | 49 (1.7)                                                               | 469 (17.5)                                                               | 23 (0.9)                                                               |
| Severe                      | 15 (0.5)                                                                 | 2 (0.1)                                                                | 69 (2.6)                                                                 | 2 (0.1)                                                                |

|                                                    | <b>COMIRNATY<br/>Dose 1<br/>N<sup>a</sup>=2899<br/>n<sup>b</sup> (%)</b> | <b>Placebo<br/>Dose 1<br/>N<sup>a</sup>=2908<br/>n<sup>b</sup> (%)</b> | <b>COMIRNATY<br/>Dose 2<br/>N<sup>a</sup>=2682<br/>n<sup>b</sup> (%)</b> | <b>Placebo<br/>Dose 2<br/>N<sup>a</sup>=2684<br/>n<sup>b</sup> (%)</b> |
|----------------------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------|
| <b>Vomiting<sup>d</sup></b>                        |                                                                          |                                                                        |                                                                          |                                                                        |
| Any                                                | 34 (1.2)                                                                 | 36 (1.2)                                                               | 58 (2.2)                                                                 | 30 (1.1)                                                               |
| Mild                                               | 29 (1.0)                                                                 | 30 (1.0)                                                               | 42 (1.6)                                                                 | 20 (0.7)                                                               |
| Moderate                                           | 5 (0.2)                                                                  | 5 (0.2)                                                                | 12 (0.4)                                                                 | 10 (0.4)                                                               |
| Severe                                             | 0                                                                        | 1 (0.0)                                                                | 4 (0.1)                                                                  | 0                                                                      |
| <b>Diarrhea<sup>e</sup></b>                        |                                                                          |                                                                        |                                                                          |                                                                        |
| Any                                                | 309 (10.7)                                                               | 323 (11.1)                                                             | 269 (10.0)                                                               | 205 (7.6)                                                              |
| Mild                                               | 251 (8.7)                                                                | 264 (9.1)                                                              | 219 (8.2)                                                                | 169 (6.3)                                                              |
| Moderate                                           | 55 (1.9)                                                                 | 58 (2.0)                                                               | 44 (1.6)                                                                 | 35 (1.3)                                                               |
| Severe                                             | 3 (0.1)                                                                  | 1 (0.0)                                                                | 6 (0.2)                                                                  | 1 (0.0)                                                                |
| <b>New or worsened muscle pain<sup>c</sup></b>     |                                                                          |                                                                        |                                                                          |                                                                        |
| Any                                                | 664 (22.9)                                                               | 329 (11.3)                                                             | 1055 (39.3)                                                              | 237 (8.8)                                                              |
| Mild                                               | 353 (12.2)                                                               | 231 (7.9)                                                              | 441 (16.4)                                                               | 150 (5.6)                                                              |
| Moderate                                           | 296 (10.2)                                                               | 96 (3.3)                                                               | 552 (20.6)                                                               | 84 (3.1)                                                               |
| Severe                                             | 15 (0.5)                                                                 | 2 (0.1)                                                                | 62 (2.3)                                                                 | 3 (0.1)                                                                |
| <b>New or worsened joint pain<sup>c</sup></b>      |                                                                          |                                                                        |                                                                          |                                                                        |
| Any                                                | 342 (11.8)                                                               | 168 (5.8)                                                              | 638 (23.8)                                                               | 147 (5.5)                                                              |
| Mild                                               | 200 (6.9)                                                                | 112 (3.9)                                                              | 291 (10.9)                                                               | 82 (3.1)                                                               |
| Moderate                                           | 137 (4.7)                                                                | 55 (1.9)                                                               | 320 (11.9)                                                               | 61 (2.3)                                                               |
| Severe                                             | 5 (0.2)                                                                  | 1 (0.0)                                                                | 27 (1.0)                                                                 | 4 (0.1)                                                                |
| Use of antipyretic or pain medication <sup>f</sup> | 805 (27.8)                                                               | 398 (13.7)                                                             | 1213 (45.2)                                                              | 320 (11.9)                                                             |

Notes: Reactions and use of antipyretic or pain medication were collected in the electronic diary (e-diary) from Day 1 to Day 7 after each dose.

No Grade 4 solicited systemic reactions were reported in participants 16 through 55 years of age.

\* Randomized participants in the safety analysis population who received at least 1 dose of the study intervention. Participants with chronic, stable HIV infection were excluded.

a. N = Number of participants reporting at least 1 yes or no response for the specified reaction after the specified dose. The N for each reaction or use of antipyretic or pain medication was the same, therefore, this information was included in the column header.

b. n = Number of participants with the specified reaction.

c. Mild: does not interfere with activity; Moderate: some interference with activity; Severe: prevents daily activity.

d. Mild: 1 to 2 times in 24 hours; Moderate: >2 times in 24 hours; Severe: requires intravenous hydration.

e. Mild: 2 to 3 loose stools in 24 hours; Moderate: 4 to 5 loose stools in 24 hours; Severe: 6 or more loose stools in 24 hours.

f. Severity was not collected for use of antipyretic or pain medication.

**Table 3: Study 2 – Frequency and Percentages of Participants with Solicited Local Reactions, by Maximum Severity, Within 7 Days After Each Dose – Participants 56 Years of Age and Older – Reactogenicity Subset of the Safety Population\***

|                            | <b>COMIRNATY<br/>Dose 1<br/>N<sup>a</sup>=2008<br/>n<sup>b</sup> (%)</b> | <b>Placebo<br/>Dose 1<br/>N<sup>a</sup>=1989<br/>n<sup>b</sup> (%)</b> | <b>COMIRNATY<br/>Dose 2<br/>N<sup>a</sup>=1860<br/>n<sup>b</sup> (%)</b> | <b>Placebo<br/>Dose 2<br/>N<sup>a</sup>=1833<br/>n<sup>b</sup> (%)</b> |
|----------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------|
| <b>Redness<sup>c</sup></b> |                                                                          |                                                                        |                                                                          |                                                                        |
| Any (>2.0 cm)              | 106 (5.3)                                                                | 20 (1.0)                                                               | 133 (7.2)                                                                | 14 (0.8)                                                               |
| Mild                       | 71 (3.5)                                                                 | 13 (0.7)                                                               | 65 (3.5)                                                                 | 10 (0.5)                                                               |
| Moderate                   | 30 (1.5)                                                                 | 5 (0.3)                                                                | 58 (3.1)                                                                 | 3 (0.2)                                                                |
| Severe                     | 5 (0.2)                                                                  | 2 (0.1)                                                                | 10 (0.5)                                                                 | 1 (0.1)                                                                |

|                                               | <b>COMIRNATY<br/>Dose 1<br/>N<sup>a</sup>=2008<br/>n<sup>b</sup> (%)</b> | <b>Placebo<br/>Dose 1<br/>N<sup>a</sup>=1989<br/>n<sup>b</sup> (%)</b> | <b>COMIRNATY<br/>Dose 2<br/>N<sup>a</sup>=1860<br/>n<sup>b</sup> (%)</b> | <b>Placebo<br/>Dose 2<br/>N<sup>a</sup>=1833<br/>n<sup>b</sup> (%)</b> |
|-----------------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------|
| <b>Swelling<sup>c</sup></b>                   |                                                                          |                                                                        |                                                                          |                                                                        |
| Any (>2.0 cm)                                 | 141 (7.0)                                                                | 23 (1.2)                                                               | 145 (7.8)                                                                | 13 (0.7)                                                               |
| Mild                                          | 87 (4.3)                                                                 | 11 (0.6)                                                               | 80 (4.3)                                                                 | 5 (0.3)                                                                |
| Moderate                                      | 52 (2.6)                                                                 | 12 (0.6)                                                               | 61 (3.3)                                                                 | 7 (0.4)                                                                |
| Severe                                        | 2 (0.1)                                                                  | 0                                                                      | 4 (0.2)                                                                  | 1 (0.1)                                                                |
| <b>Pain at the injection site<sup>d</sup></b> |                                                                          |                                                                        |                                                                          |                                                                        |
| Any (>2.0 cm)                                 | 1408 (70.1)                                                              | 185 (9.3)                                                              | 1230 (66.1)                                                              | 143 (7.8)                                                              |
| Mild                                          | 1108 (55.2)                                                              | 177 (8.9)                                                              | 873 (46.9)                                                               | 138 (7.5)                                                              |
| Moderate                                      | 296 (14.7)                                                               | 8 (0.4)                                                                | 347 (18.7)                                                               | 5 (0.3)                                                                |
| Severe                                        | 4 (0.2)                                                                  | 0                                                                      | 10 (0.5)                                                                 | 0                                                                      |

Notes: Reactions were collected in the electronic diary (e-diary) from Day 1 to Day 7 after vaccination.

No Grade 4 solicited local reactions were reported in participants 56 years of age and older.

\* Randomized participants in the safety analysis population who received at least 1 dose of the study intervention. Participants with chronic, stable HIV infection were excluded.

a. N = Number of participants reporting at least 1 yes or no response for the specified reaction after the specified dose. The N for each reaction was the same, therefore, the information was included in the column header.

b. n = Number of participants with the specified reaction.

c. Mild: >2.0 to ≤5.0 cm; Moderate: >5.0 to ≤10.0 cm; Severe: >10.0 cm.

d. Mild: does not interfere with activity; Moderate: interferes with activity; Severe: prevents daily activity.

**Table 4: Study 2 – Frequency and Percentages of Participants with Solicited Systemic Reactions, by Maximum Severity, Within 7 Days After Each Dose – Participants 56 Years of Age and Older – Reactogenicity Subset of the Safety Population\***

|                             | <b>COMIRNATY<br/>Dose 1<br/>N<sup>a</sup>=2008<br/>n<sup>b</sup> (%)</b> | <b>Placebo<br/>Dose 1<br/>N<sup>a</sup>=1989<br/>n<sup>b</sup> (%)</b> | <b>COMIRNATY<br/>Dose 2<br/>N<sup>a</sup>=1860<br/>n<sup>b</sup> (%)</b> | <b>Placebo<br/>Dose 2<br/>N<sup>a</sup>=1833<br/>n<sup>b</sup> (%)</b> |
|-----------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------|
| <b>Fever</b>                |                                                                          |                                                                        |                                                                          |                                                                        |
| ≥38.0°C                     | 26 (1.3)                                                                 | 8 (0.4)                                                                | 219 (11.8)                                                               | 4 (0.2)                                                                |
| ≥38.0°C to 38.4°C           | 23 (1.1)                                                                 | 3 (0.2)                                                                | 158 (8.5)                                                                | 2 (0.1)                                                                |
| >38.4°C to 38.9°C           | 2 (0.1)                                                                  | 3 (0.2)                                                                | 54 (2.9)                                                                 | 1 (0.1)                                                                |
| >38.9°C to 40.0°C           | 1 (0.0)                                                                  | 2 (0.1)                                                                | 7 (0.4)                                                                  | 1 (0.1)                                                                |
| >40.0°C                     | 0                                                                        | 0                                                                      | 0                                                                        | 0                                                                      |
| <b>Fatigue<sup>c</sup></b>  |                                                                          |                                                                        |                                                                          |                                                                        |
| Any                         | 677 (33.7)                                                               | 447 (22.5)                                                             | 949 (51.0)                                                               | 306 (16.7)                                                             |
| Mild                        | 415 (20.7)                                                               | 281 (14.1)                                                             | 391 (21.0)                                                               | 183 (10.0)                                                             |
| Moderate                    | 259 (12.9)                                                               | 163 (8.2)                                                              | 497 (26.7)                                                               | 121 (6.6)                                                              |
| Severe                      | 3 (0.1)                                                                  | 3 (0.2)                                                                | 60 (3.2)                                                                 | 2 (0.1)                                                                |
| Grade 4                     | 0                                                                        | 0                                                                      | 1 (0.1)                                                                  | 0                                                                      |
| <b>Headache<sup>c</sup></b> |                                                                          |                                                                        |                                                                          |                                                                        |
| Any                         | 503 (25.0)                                                               | 363 (18.3)                                                             | 733 (39.4)                                                               | 259 (14.1)                                                             |
| Mild                        | 381 (19.0)                                                               | 267 (13.4)                                                             | 464 (24.9)                                                               | 189 (10.3)                                                             |
| Moderate                    | 120 (6.0)                                                                | 93 (4.7)                                                               | 256 (13.8)                                                               | 65 (3.5)                                                               |
| Severe                      | 2 (0.1)                                                                  | 3 (0.2)                                                                | 13 (0.7)                                                                 | 5 (0.3)                                                                |



|                                                              | <b>COMIRNATY<br/>Dose 1<br/>N<sup>a</sup>=2008<br/>n<sup>b</sup> (%)</b> | <b>Placebo<br/>Dose 1<br/>N<sup>a</sup>=1989<br/>n<sup>b</sup> (%)</b> | <b>COMIRNATY<br/>Dose 2<br/>N<sup>a</sup>=1860<br/>n<sup>b</sup> (%)</b> | <b>Placebo<br/>Dose 2<br/>N<sup>a</sup>=1833<br/>n<sup>b</sup> (%)</b> |
|--------------------------------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------|
| <b>Chills<sup>c</sup></b>                                    |                                                                          |                                                                        |                                                                          |                                                                        |
| Any                                                          | 130 (6.5)                                                                | 69 (3.5)                                                               | 435 (23.4)                                                               | 57 (3.1)                                                               |
| Mild                                                         | 102 (5.1)                                                                | 49 (2.5)                                                               | 229 (12.3)                                                               | 45 (2.5)                                                               |
| Moderate                                                     | 28 (1.4)                                                                 | 19 (1.0)                                                               | 185 (9.9)                                                                | 12 (0.7)                                                               |
| Severe                                                       | 0                                                                        | 1 (0.1)                                                                | 21 (1.1)                                                                 | 0                                                                      |
| <b>Vomiting<sup>d</sup></b>                                  |                                                                          |                                                                        |                                                                          |                                                                        |
| Any                                                          | 10 (0.5)                                                                 | 9 (0.5)                                                                | 13 (0.7)                                                                 | 5 (0.3)                                                                |
| Mild                                                         | 9 (0.4)                                                                  | 9 (0.5)                                                                | 10 (0.5)                                                                 | 5 (0.3)                                                                |
| Moderate                                                     | 1 (0.0)                                                                  | 0                                                                      | 1 (0.1)                                                                  | 0                                                                      |
| Severe                                                       | 0                                                                        | 0                                                                      | 2 (0.1)                                                                  | 0                                                                      |
| <b>Diarrhea<sup>e</sup></b>                                  |                                                                          |                                                                        |                                                                          |                                                                        |
| Any                                                          | 168 (8.4)                                                                | 130 (6.5)                                                              | 152 (8.2)                                                                | 102 (5.6)                                                              |
| Mild                                                         | 137 (6.8)                                                                | 109 (5.5)                                                              | 125 (6.7)                                                                | 76 (4.1)                                                               |
| Moderate                                                     | 27 (1.3)                                                                 | 20 (1.0)                                                               | 25 (1.3)                                                                 | 22 (1.2)                                                               |
| Severe                                                       | 4 (0.2)                                                                  | 1 (0.1)                                                                | 2 (0.1)                                                                  | 4 (0.2)                                                                |
| <b>New or worsened muscle pain<sup>c</sup></b>               |                                                                          |                                                                        |                                                                          |                                                                        |
| Any                                                          | 274 (13.6)                                                               | 165 (8.3)                                                              | 537 (28.9)                                                               | 99 (5.4)                                                               |
| Mild                                                         | 183 (9.1)                                                                | 111 (5.6)                                                              | 229 (12.3)                                                               | 65 (3.5)                                                               |
| Moderate                                                     | 90 (4.5)                                                                 | 51 (2.6)                                                               | 288 (15.5)                                                               | 33 (1.8)                                                               |
| Severe                                                       | 1 (0.0)                                                                  | 3 (0.2)                                                                | 20 (1.1)                                                                 | 1 (0.1)                                                                |
| <b>New or worsened joint pain<sup>c</sup></b>                |                                                                          |                                                                        |                                                                          |                                                                        |
| Any                                                          | 175 (8.7)                                                                | 124 (6.2)                                                              | 353 (19.0)                                                               | 72 (3.9)                                                               |
| Mild                                                         | 119 (5.9)                                                                | 78 (3.9)                                                               | 183 (9.8)                                                                | 44 (2.4)                                                               |
| Moderate                                                     | 53 (2.6)                                                                 | 45 (2.3)                                                               | 161 (8.7)                                                                | 27 (1.5)                                                               |
| Severe                                                       | 3 (0.1)                                                                  | 1 (0.1)                                                                | 9 (0.5)                                                                  | 1 (0.1)                                                                |
| <b>Use of antipyretic or<br/>pain medication<sup>f</sup></b> | 382 (19.0)                                                               | 224 (11.3)                                                             | 688 (37.0)                                                               | 170 (9.3)                                                              |

Notes: Reactions and use of antipyretic or pain medication were collected in the electronic diary (e-diary) from Day 1 to Day 7 after each dose.

The only Grade 4 solicited systemic reaction reported in participants 56 years of age and older was fatigue.

\* Randomized participants in the safety analysis population who received at least 1 dose of the study intervention. Participants with chronic, stable HIV infection were excluded.

a. N = Number of participants reporting at least 1 yes or no response for the specified reaction after the specified dose. N for each reaction or use of antipyretic or pain medication was the same, therefore was included in the column header.

b. n = Number of participants with the specified reaction.

c. Mild: does not interfere with activity; Moderate: some interference with activity; Severe: prevents daily activity; Grade 4 reactions were defined in the clinical study protocol as emergency room visit or hospitalization for severe fatigue, severe headache, severe chills, severe muscle pain, or severe joint pain.

d. Mild: 1 to 2 times in 24 hours; Moderate: >2 times in 24 hours; Severe: requires intravenous hydration; Grade 4 emergency visit or hospitalization for severe vomiting.

e. Mild: 2 to 3 loose stools in 24 hours; Moderate: 4 to 5 loose stools in 24 hours; Severe: 6 or more loose stools in 24 hours; Grade 4: emergency room or hospitalization for severe diarrhea.

f. Severity was not collected for use of antipyretic or pain medication.

In participants with chronic, stable HIV infection the frequencies of solicited local and systemic adverse reactions were similar to or lower than those observed for all participants 16 years of age and older.

### *Unsolicited Adverse Events*

Overall, 11,253 (51.1%) participants in the COMIRNATY group and 11,316 (51.4%) participants in the placebo group had follow-up time between  $\geq 4$  months to  $< 6$  months after Dose 2 in the blinded placebo-controlled follow-up period with an additional 1,778 (8.1%) and 1,304 (5.9%) with  $\geq 6$  months of blinded follow-up time in the COMIRNATY and placebo groups, respectively.

A total of 12,006 (54.5%) participants originally randomized to COMIRNATY had  $\geq 6$  months total (blinded and unblinded) follow-up after Dose 2.

In an analysis of all unsolicited adverse events reported following any dose, through 1 month after Dose 2, in participants 16 years of age and older (N=43,847; 21,926 COMIRNATY group vs. 21,921 placebo group), those assessed as adverse reactions not already captured by solicited local and systemic reactions were nausea (274 vs. 87), malaise (130 vs. 22), lymphadenopathy (83 vs. 7), asthenia (76 vs. 25), decreased appetite (39 vs. 9), hyperhidrosis (31 vs. 9), lethargy (25 vs. 6), and night sweats (17 vs. 3).

In analyses of all unsolicited adverse events in Study 2 from Dose 1 up to the participant unblinding date, 58.2% of study participants had at least 4 months of follow-up after Dose 2. Among participants 16 through 55 years of age who received at least 1 dose of study vaccine, 12,995 of whom received COMIRNATY and 13,026 of whom received placebo, unsolicited adverse events were reported by 4,396 (33.8%) participants in the COMIRNATY group and 2,136 (16.4%) participants in the placebo group. In a similar analysis in participants 56 years of age and older that included 8,931 COMIRNATY recipients and 8,895 placebo recipients, unsolicited adverse events were reported by 2,551 (28.6%) participants in the COMIRNATY group and 1,432 (16.1%) participants in the placebo group. Among participants with confirmed stable HIV infection that included 100 COMIRNATY recipients and 100 placebo recipients, unsolicited adverse events were reported by 29 (29%) participants in the COMIRNATY group and 15 (15%) participants in the placebo group. The higher frequency of reported unsolicited adverse events among COMIRNATY recipients compared to placebo recipients was primarily attributed to events that are consistent with adverse reactions solicited among participants in the reactogenicity subset (Table 3 and Table 4).

Throughout the placebo-controlled safety follow-up period, Bell's palsy (facial paralysis) was reported by 4 participants in the COMIRNATY group and 2 participants in the placebo group. Onset of facial paralysis was Day 37 after Dose 1 (participant did not receive Dose 2) and Days 3, 9, and 48 after Dose 2. In the placebo group the onset of facial paralysis was Day 32 and Day 102. Currently available information is insufficient to determine a causal relationship with the vaccine. In the analysis of blinded, placebo-controlled follow-up, there were no other notable patterns or numerical imbalances between treatment groups for specific categories of non-serious adverse events (including other neurologic or neuro-inflammatory, and thrombotic events) that would suggest a causal relationship to COMIRNATY. In the analysis of unblinded follow-up, there were no notable patterns of specific categories of non-serious adverse events that would suggest a causal relationship to COMIRNATY.

### *Serious Adverse Events*

In Study 2, among participants 16 through 55 years of age who had received at least 1 dose of vaccine or placebo (COMIRNATY = 12,995; placebo = 13,026), serious adverse events from Dose 1 up to the participant unblinding date in ongoing follow-up were reported by 103 (0.8%) COMIRNATY recipients and 117 (0.9%) placebo recipients. In a similar analysis, in participants 56 years of age and older (COMIRNATY = 8,931; placebo = 8,895), serious adverse events were reported by 165 (1.8%) COMIRNATY recipients and 151 (1.7%) placebo recipients who received at least 1 dose of COMIRNATY or placebo, respectively. In these analyses, 58.2% of study participants had at least 4 months of follow-up after Dose 2. Among participants with confirmed



stable HIV infection serious adverse events from Dose 1 up to the participant unblinding date in ongoing follow-up were reported by 2 (2%) COMIRNATY recipients and 2 (2%) placebo recipients.

In the analysis of blinded, placebo-controlled follow-up, there were no notable patterns between treatment groups for specific categories of serious adverse events (including neurologic, neuro-inflammatory, and thrombotic events) that would suggest a causal relationship to COMIRNATY. In the analysis of unblinded follow-up, there were no notable patterns of specific categories of serious adverse events that would suggest a causal relationship to COMIRNATY.

### Adolescents 12 Through 15 Years of Age

In Study 2, 2,260 adolescents (1,131 COMIRNATY; 1,129 placebo) were 12 through 15 years of age. At the time of the analysis of the ongoing Study 2 with a data cutoff of September 2, 2021, there were 1,559 (69.0%) adolescents (786 COMIRNATY and 773 placebo) 12 through 15 years of age followed for  $\geq 4$  months after the second dose. The safety evaluation in Study 2 is ongoing.

Demographic characteristics in Study 2 were generally similar with regard to age, gender, race, and ethnicity among adolescents who received COMIRNATY and those who received placebo. Overall, among the adolescents who received COMIRNATY, 50.1% were male and 49.9% were female, 85.8% were White, 4.6% were Black or African American, 11.7% were Hispanic/Latino, 6.4% were Asian, and 0.4% were American Indian/Alaska Native.

### *Local and Systemic Adverse Reactions Solicited in Study 2*

In adolescents 12 through 15 years of age after receiving Dose 2, the mean duration of pain at the injection site was 2.5 days (range 1 to 11 days), for redness 1.8 days (range 1 to 5 days), and for swelling 1.6 days (range 1 to 5 days) in the COMIRNATY group.

**Table 5: Study 2 – Frequency and Percentages of Adolescents With Solicited Local Reactions, by Maximum Severity, Within 7 Days After Each Dose – Adolescents 12 Through 15 Years of Age – Safety Population\***

|                             | <b>COMIRNATY<br/>Dose 1<br/>N<sup>a</sup>=1127<br/>n<sup>b</sup> (%)</b> | <b>Placebo<br/>Dose 1<br/>N<sup>a</sup>=1127<br/>n<sup>b</sup> (%)</b> | <b>COMIRNATY<br/>Dose 2<br/>N<sup>a</sup>=1097<br/>n<sup>b</sup> (%)</b> | <b>Placebo<br/>Dose 2<br/>N<sup>a</sup>=1078<br/>n<sup>b</sup> (%)</b> |
|-----------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------|
| <b>Redness<sup>c</sup></b>  |                                                                          |                                                                        |                                                                          |                                                                        |
| Any (>2 cm)                 | 65 (5.8)                                                                 | 12 (1.1)                                                               | 55 (5.0)                                                                 | 10 (0.9)                                                               |
| Mild                        | 44 (3.9)                                                                 | 11 (1.0)                                                               | 29 (2.6)                                                                 | 8 (0.7)                                                                |
| Moderate                    | 20 (1.8)                                                                 | 1 (0.1)                                                                | 26 (2.4)                                                                 | 2 (0.2)                                                                |
| Severe                      | 1 (0.1)                                                                  | 0 (0.0)                                                                | 0 (0.0)                                                                  | 0 (0.0)                                                                |
| <b>Swelling<sup>c</sup></b> |                                                                          |                                                                        |                                                                          |                                                                        |
| Any (>2 cm)                 | 78 (6.9)                                                                 | 11 (1.0)                                                               | 54 (4.9)                                                                 | 6 (0.6)                                                                |
| Mild                        | 55 (4.9)                                                                 | 9 (0.8)                                                                | 36 (3.3)                                                                 | 4 (0.4)                                                                |
| Moderate                    | 23 (2.0)                                                                 | 2 (0.2)                                                                | 18 (1.6)                                                                 | 2 (0.2)                                                                |
| Severe                      | 0 (0.0)                                                                  | 0 (0.0)                                                                | 0 (0.0)                                                                  | 0 (0.0)                                                                |

|                                               | <b>COMIRNATY<br/>Dose 1<br/>N<sup>a</sup>=1127<br/>n<sup>b</sup> (%)</b> | <b>Placebo<br/>Dose 1<br/>N<sup>a</sup>=1127<br/>n<sup>b</sup> (%)</b> | <b>COMIRNATY<br/>Dose 2<br/>N<sup>a</sup>=1097<br/>n<sup>b</sup> (%)</b> | <b>Placebo<br/>Dose 2<br/>N<sup>a</sup>=1078<br/>n<sup>b</sup> (%)</b> |
|-----------------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------|
| <b>Pain at the injection site<sup>d</sup></b> |                                                                          |                                                                        |                                                                          |                                                                        |
| Any                                           | 971 (86.2)                                                               | 263 (23.3)                                                             | 866 (78.9)                                                               | 193 (17.9)                                                             |
| Mild                                          | 467 (41.4)                                                               | 227 (20.1)                                                             | 466 (42.5)                                                               | 164 (15.2)                                                             |
| Moderate                                      | 493 (43.7)                                                               | 36 (3.2)                                                               | 393 (35.8)                                                               | 29 (2.7)                                                               |
| Severe                                        | 11 (1.0)                                                                 | 0 (0.0)                                                                | 7 (0.6)                                                                  | 0 (0.0)                                                                |

Note: Reactions were collected in the electronic diary (e-diary) from Day 1 to Day 7 after vaccination.

\* Randomized participants in the safety analysis population who received at least 1 dose of the study intervention.

a. N = Number of participants reporting at least 1 yes or no response for the specified reaction after the specified dose.

b. n = Number of participants with the specified reaction.

c. Mild: >2.0 to ≤5.0 cm; Moderate: >5.0 to ≤10.0 cm; Severe: >10.0 cm.

d. Mild: does not interfere with activity; Moderate: interferes with activity; Severe: prevents daily activity.

**Table 6: Study 2 – Frequency and Percentages of Adolescents with Solicited Systemic Reactions, by Maximum Severity, Within 7 Days After Each Dose – Adolescents 12 Through 15 Years of Age – Safety Population\***

|                             | <b>COMIRNATY<br/>Dose 1<br/>N<sup>a</sup>=1127<br/>n<sup>b</sup> (%)</b> | <b>Placebo<br/>Dose 1<br/>N<sup>a</sup>=1127<br/>n<sup>b</sup> (%)</b> | <b>COMIRNATY<br/>Dose 2<br/>N<sup>a</sup>=1097<br/>n<sup>b</sup> (%)</b> | <b>Placebo<br/>Dose 2<br/>N<sup>a</sup>=1078<br/>n<sup>b</sup> (%)</b> |
|-----------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------|
| <b>Fever</b>                |                                                                          |                                                                        |                                                                          |                                                                        |
| ≥38.0°C                     | 114 (10.1)                                                               | 12 (1.1)                                                               | 215 (19.6)                                                               | 7 (0.6)                                                                |
| ≥38.0°C to 38.4°C           | 74 (6.6)                                                                 | 8 (0.7)                                                                | 107 (9.8)                                                                | 5 (0.5)                                                                |
| >38.4°C to 38.9°C           | 29 (2.6)                                                                 | 2 (0.2)                                                                | 83 (7.6)                                                                 | 1 (0.1)                                                                |
| >38.9°C to 40.0°C           | 10 (0.9)                                                                 | 2 (0.2)                                                                | 25 (2.3)                                                                 | 1 (0.1)                                                                |
| >40.0°C                     | 1 (0.1)                                                                  | 0 (0.0)                                                                | 0 (0.0)                                                                  | 0 (0.0)                                                                |
| <b>Fatigue<sup>c</sup></b>  |                                                                          |                                                                        |                                                                          |                                                                        |
| Any                         | 677 (60.1)                                                               | 457 (40.6)                                                             | 726 (66.2)                                                               | 264 (24.5)                                                             |
| Mild                        | 278 (24.7)                                                               | 250 (22.2)                                                             | 232 (21.1)                                                               | 133 (12.3)                                                             |
| Moderate                    | 384 (34.1)                                                               | 199 (17.7)                                                             | 468 (42.7)                                                               | 127 (11.8)                                                             |
| Severe                      | 15 (1.3)                                                                 | 8 (0.7)                                                                | 26 (2.4)                                                                 | 4 (0.4)                                                                |
| <b>Headache<sup>c</sup></b> |                                                                          |                                                                        |                                                                          |                                                                        |
| Any                         | 623 (55.3)                                                               | 396 (35.1)                                                             | 708 (64.5)                                                               | 264 (24.5)                                                             |
| Mild                        | 361 (32.0)                                                               | 256 (22.7)                                                             | 302 (27.5)                                                               | 170 (15.8)                                                             |
| Moderate                    | 251 (22.3)                                                               | 131 (11.6)                                                             | 384 (35.0)                                                               | 93 (8.6)                                                               |
| Severe                      | 11 (1.0)                                                                 | 9 (0.8)                                                                | 22 (2.0)                                                                 | 1 (0.1)                                                                |
| <b>Chills<sup>c</sup></b>   |                                                                          |                                                                        |                                                                          |                                                                        |
| Any                         | 311 (27.6)                                                               | 109 (9.7)                                                              | 455 (41.5)                                                               | 74 (6.9)                                                               |
| Mild                        | 195 (17.3)                                                               | 82 (7.3)                                                               | 221 (20.1)                                                               | 53 (4.9)                                                               |
| Moderate                    | 111 (9.8)                                                                | 25 (2.2)                                                               | 214 (19.5)                                                               | 21 (1.9)                                                               |
| Severe                      | 5 (0.4)                                                                  | 2 (0.2)                                                                | 20 (1.8)                                                                 | 0 (0.0)                                                                |
| <b>Vomiting<sup>d</sup></b> |                                                                          |                                                                        |                                                                          |                                                                        |
| Any                         | 31 (2.8)                                                                 | 10 (0.9)                                                               | 29 (2.6)                                                                 | 12 (1.1)                                                               |
| Mild                        | 30 (2.7)                                                                 | 8 (0.7)                                                                | 25 (2.3)                                                                 | 11 (1.0)                                                               |
| Moderate                    | 0 (0.0)                                                                  | 2 (0.2)                                                                | 4 (0.4)                                                                  | 1 (0.1)                                                                |
| Severe                      | 1 (0.1)                                                                  | 0 (0.0)                                                                | 0 (0.0)                                                                  | 0 (0.0)                                                                |

|                                                    | <b>COMIRNATY<br/>Dose 1<br/>N<sup>a</sup>=1127<br/>n<sup>b</sup> (%)</b> | <b>Placebo<br/>Dose 1<br/>N<sup>a</sup>=1127<br/>n<sup>b</sup> (%)</b> | <b>COMIRNATY<br/>Dose 2<br/>N<sup>a</sup>=1097<br/>n<sup>b</sup> (%)</b> | <b>Placebo<br/>Dose 2<br/>N<sup>a</sup>=1078<br/>n<sup>b</sup> (%)</b> |
|----------------------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------|
| <b>Diarrhea<sup>c</sup></b>                        |                                                                          |                                                                        |                                                                          |                                                                        |
| Any                                                | 90 (8.0)                                                                 | 82 (7.3)                                                               | 65 (5.9)                                                                 | 44 (4.1)                                                               |
| Mild                                               | 77 (6.8)                                                                 | 72 (6.4)                                                               | 59 (5.4)                                                                 | 39 (3.6)                                                               |
| Moderate                                           | 13 (1.2)                                                                 | 10 (0.9)                                                               | 6 (0.5)                                                                  | 5 (0.5)                                                                |
| Severe                                             | 0 (0.0)                                                                  | 0 (0.0)                                                                | 0 (0.0)                                                                  | 0 (0.0)                                                                |
| <b>New or worsened muscle pain<sup>c</sup></b>     |                                                                          |                                                                        |                                                                          |                                                                        |
| Any                                                | 272 (24.1)                                                               | 148 (13.1)                                                             | 355 (32.4)                                                               | 90 (8.3)                                                               |
| Mild                                               | 125 (11.1)                                                               | 88 (7.8)                                                               | 152 (13.9)                                                               | 51 (4.7)                                                               |
| Moderate                                           | 145 (12.9)                                                               | 60 (5.3)                                                               | 197 (18.0)                                                               | 37 (3.4)                                                               |
| Severe                                             | 2 (0.2)                                                                  | 0 (0.0)                                                                | 6 (0.5)                                                                  | 2 (0.2)                                                                |
| <b>New or worsened joint pain<sup>c</sup></b>      |                                                                          |                                                                        |                                                                          |                                                                        |
| Any                                                | 109 (9.7)                                                                | 77 (6.8)                                                               | 173 (15.8)                                                               | 51 (4.7)                                                               |
| Mild                                               | 66 (5.9)                                                                 | 50 (4.4)                                                               | 91 (8.3)                                                                 | 30 (2.8)                                                               |
| Moderate                                           | 42 (3.7)                                                                 | 27 (2.4)                                                               | 78 (7.1)                                                                 | 21 (1.9)                                                               |
| Severe                                             | 1 (0.1)                                                                  | 0 (0.0)                                                                | 4 (0.4)                                                                  | 0 (0.0)                                                                |
| Use of antipyretic or pain medication <sup>f</sup> | 413 (36.6)                                                               | 111 (9.8)                                                              | 557 (50.8)                                                               | 95 (8.8)                                                               |

Note: Events and use of antipyretic or pain medication were collected in the electronic diary (e-diary) from Day 1 to Day 7 after each dose.

\* Randomized participants in the safety analysis population who received at least 1 dose of the study intervention.

a. N = Number of participants reporting at least 1 yes or no response for the specified event after the specified dose.

b. n = Number of participants with the specified reaction.

c. Mild: does not interfere with activity; Moderate: some interference with activity; Severe: prevents daily activity.

d. Mild: 1 to 2 times in 24 hours; Moderate: >2 times in 24 hours; Severe: requires intravenous hydration.

e. Mild: 2 to 3 loose stools in 24 hours; Moderate: 4 to 5 loose stools in 24 hours; Severe: 6 or more loose stools in 24 hours.

f. Severity was not collected for use of antipyretic or pain medication.

### *Unsolicited Adverse Events*

In Study 2, 2,260 adolescents (1,131 COMIRNATY; 1,129 placebo) were 12 through 15 years of age. Of these, 634 (56.1%) participants in the COMIRNATY group and 629 (55.7%) participants in the placebo group had follow-up time between  $\geq 4$  months to <6 months after Dose 2 in the blinded placebo-controlled follow-up period with an additional 152 (13.4%) and 144 (12.8%) with  $\geq 6$  months of blinded follow-up time in the COMIRNATY and placebo groups, respectively.

A total of 1,113 (98.4%) participants 12 through 15 years of age originally randomized to COMIRNATY had  $\geq 6$  months total (blinded and unblinded) follow-up after Dose 2.

An analysis of all unsolicited adverse events in Study 2 from Dose 1 up to the participant unblinding date was conducted. Among participants 12 through 15 years of age who received at least one dose of study vaccine, unsolicited adverse events were reported by 95 (8.4%) participants in the COMIRNATY group and 113 (10.0%) participants in the placebo group.

In an analysis of all unsolicited adverse events reported during blinded follow-up from Dose 1 through 1 month after Dose 2, in adolescents 12 to 15 years of age, those assessed as adverse reactions not already captured by solicited local and systemic reactions were lymphadenopathy (9 vs. 2), and nausea (5 vs. 2).

In the analysis of blinded, placebo-controlled follow-up, there were no other notable patterns or numerical imbalances between treatment groups for specific categories of unsolicited adverse events (including other neurologic or neuro-inflammatory, and thrombotic events) that would suggest a causal relationship to COMIRNATY. In the analysis of unblinded follow-up, there were no notable patterns of specific categories of non-serious adverse events that would suggest a causal relationship to COMIRNATY.

#### *Serious Adverse Events*

In Study 2, among participants 12 through 15 years of age who had received at least 1 dose of vaccine or placebo (COMIRNATY = 1,131; placebo = 1,129), serious adverse events from Dose 1 up to the participant unblinding date in ongoing follow-up were reported by 10 (0.9%) COMIRNATY recipients and 2 (0.2%) placebo recipients. In these analyses, 69.0% of study participants had at least 4 months of follow-up after Dose 2. In the analysis of blinded, placebo-controlled follow-up, there were no notable patterns between treatment groups for specific categories of serious adverse events (including neurologic, neuro-inflammatory, and thrombotic events) that would suggest a causal relationship to COMIRNATY. In the analysis of unblinded follow-up, there were no notable patterns of specific categories of serious adverse events that would suggest a causal relationship to COMIRNATY.

## **6.2 Postmarketing Experience**

The following adverse reactions have been identified during postmarketing use of COMIRNATY, including under Emergency Use Authorization. Because these reactions are reported voluntarily from a population of uncertain size, it is not always possible to reliably estimate their frequency or establish a causal relationship to vaccine exposure.

Cardiac Disorders: myocarditis, pericarditis

Gastrointestinal Disorders: diarrhea, vomiting

Immune System Disorders: severe allergic reactions, including anaphylaxis, and other hypersensitivity reactions (e.g., rash, pruritus, urticaria, angioedema)

Musculoskeletal and Connective Tissue Disorders: pain in extremity (arm)

## **8 USE IN SPECIFIC POPULATIONS**

### **8.1 Pregnancy**

There is a pregnancy exposure registry that monitors pregnancy outcomes in women exposed to COMIRNATY during pregnancy. Women who are vaccinated with COMIRNATY during pregnancy are encouraged to enroll in the registry by visiting <https://mothertobaby.org/ongoing-study/covid19-vaccines/>.

#### Risk Summary

All pregnancies have a risk of birth defect, loss, or other adverse outcomes. In the US general population, the estimated background risk of major birth defects and miscarriage in clinically recognized pregnancies is 2% to 4% and 15% to 20%, respectively. Available data on COMIRNATY administered to pregnant women are insufficient to inform vaccine-associated risks in pregnancy.

A developmental toxicity study has been performed in female rats administered the equivalent of a single human dose of COMIRNATY on 4 occasions, twice prior to mating and twice during gestation. These studies revealed no evidence of harm to the fetus due to the vaccine (*see Animal Data*).

## Data

### *Animal Data*

In a developmental toxicity study, 0.06 mL of a vaccine formulation containing the same quantity of nucleoside-modified messenger ribonucleic acid (mRNA) (30 mcg) and other ingredients included in a single human dose of COMIRNATY was administered to female rats by the intramuscular route on 4 occasions: 21 and 14 days prior to mating, and on gestation days 9 and 20. No vaccine-related adverse effects on female fertility, fetal development, or postnatal development were reported in the study.

## **8.2 Lactation**

### Risk Summary

It is not known whether COMIRNATY is excreted in human milk. Data are not available to assess the effects of COMIRNATY on the breastfed infant or on milk production/excretion. The developmental and health benefits of breastfeeding should be considered along with the mother's clinical need for COMIRNATY and any potential adverse effects on the breastfed child from COMIRNATY or from the underlying maternal condition. For preventive vaccines, the underlying maternal condition is susceptibility to disease prevented by the vaccine.

## **8.4 Pediatric Use**

Safety and effectiveness of COMIRNATY in individuals 12 through 17 years of age is based on safety and effectiveness data in this age group and in adults [*see Adverse Reactions (6) and Clinical Studies (14.1)*].

The safety and effectiveness of COMIRNATY in individuals younger than 12 years of age have not been established.

## **8.5 Geriatric Use**

Of the total number of COMIRNATY recipients in Study 2 as of March 13, 2021 (N = 22,026), 20.7% (n = 4,552) were 65 years of age and older and 4.2% (n = 925) were 75 years of age and older [*see Clinical Studies (14.1)*]. No overall differences in safety or effectiveness were observed between these recipients and younger recipients.

## **11 DESCRIPTION**

COMIRNATY (COVID-19 Vaccine, mRNA) is a sterile suspension for injection for intramuscular use. COMIRNATY is supplied as a frozen suspension in multiple dose vials with purple caps and labels with purple borders; each vial must be diluted with 1.8 mL of sterile 0.9% Sodium Chloride Injection, USP prior to use to form the vaccine. Each 0.3 mL dose of COMIRNATY supplied in multiple dose vials with purple caps and labels with purple borders contains 30 mcg of a nucleoside-modified messenger RNA (mRNA) encoding the viral spike (S) glycoprotein of SARS-CoV-2.

Each 0.3 mL dose of the COMIRNATY supplied in multiple dose vials with purple caps and labels with purple borders also includes the following ingredients:

lipids (0.43 mg ((4-hydroxybutyl)azanediyl)bis(hexane-6,1-diyl)bis(2-hexyldecanoate), 0.05 mg 2-(polyethylene glycol 2000)-N,N-ditetradecylacetamide, 0.09 mg 1,2-distearoyl-sn-glycero-3-phosphocholine, and 0.2 mg cholesterol), 0.01 mg potassium chloride, 0.01 mg monobasic potassium phosphate, 0.36 mg sodium chloride, 0.07 mg dibasic sodium phosphate

dihydrate, and 6 mg sucrose. The diluent (sterile 0.9% Sodium Chloride Injection, USP) contributes an additional 2.16 mg sodium chloride per dose.

COMIRNATY does not contain preservative.

The vial stoppers are not made with natural rubber latex.

## **12 CLINICAL PHARMACOLOGY**

### **12.1 Mechanism of Action**

The nucleoside-modified mRNA in COMIRNATY is formulated in lipid particles, which enable delivery of the mRNA into host cells to allow expression of the SARS-CoV-2 S antigen. The vaccine elicits an immune response to the S antigen, which protects against COVID-19.

## **13 NONCLINICAL TOXICOLOGY**

### **13.1 Carcinogenesis, Mutagenesis, Impairment of Fertility**

COMIRNATY has not been evaluated for the potential to cause carcinogenicity, genotoxicity, or impairment of male fertility. In a developmental toxicity study in rats with COMIRNATY there were no vaccine-related effects on female fertility [*see Use in Specific Populations (8.1)*].

## **14 CLINICAL STUDIES**

### **14.1 Efficacy in Participants 16 Years of Age and Older**

Study 2 is an ongoing, multicenter, multinational, randomized, placebo-controlled, observer-blind, dose-finding, vaccine candidate–selection, and efficacy study in participants 12 years of age and older. Randomization was stratified by age: 12 through 15 years of age, 16 through 55 years of age, or 56 years of age and older, with a minimum of 40% of participants in the  $\geq 56$ -year stratum. The study excluded participants who were immunocompromised and those who had previous clinical or microbiological diagnosis of COVID-19. Participants with preexisting stable disease, defined as disease not requiring significant change in therapy or hospitalization for worsening disease during the 6 weeks before enrollment, were included as were participants with known stable infection with HIV, hepatitis C virus (HCV), or hepatitis B virus (HBV).

In Study 2, based on data accrued through March 13, 2021, approximately 44,000 participants 12 years of age and older were randomized equally and received 2 doses of COMIRNATY or placebo. Participants are planned to be followed for up to 24 months, for assessments of safety and efficacy against COVID-19.

Overall, among the total participants who received COMIRNATY or placebo, 51.4% or 50.3% were male and 48.6% or 49.7% were female, 79.1% or 79.2% were 16 through 64 years of age, 20.9% or 20.8% were 65 years of age and older, 81.9% or 82.1% were White, 9.5% or 9.6% were Black or African American, 1.0% or 0.9% were American Indian or Alaska Native, 4.4% or 4.3% were Asian, 0.3% or 0.2% Native Hawaiian or other Pacific Islander, 25.6% or 25.4% were Hispanic/Latino, 73.9% or 74.1% were non-Hispanic/Latino, 0.5% or 0.5% did not report ethnicity, 46.0% or 45.7% had comorbidities [participants who have 1 or more comorbidities that increase the risk of severe COVID-19 disease: defined as subjects who had at least 1 of the Charlson comorbidity index category or body mass index (BMI)  $\geq 30$  kg/m<sup>2</sup>], respectively. The mean age at vaccination was 49.8 or 49.7 years and median age was 51.0 or 51.0 in participants who received COMIRNATY or placebo, respectively.



## Efficacy Against COVID-19

The population for the analysis of the protocol pre-specified primary efficacy endpoint included 36,621 participants 12 years of age and older (18,242 in the COMIRNATY group and 18,379 in the placebo group) who did not have evidence of prior infection with SARS-CoV-2 through 7 days after the second dose. The population in the protocol pre-specified primary efficacy analysis included all participants 12 years of age and older who had been enrolled from July 27, 2020, and followed for the development of COVID-19 through November 14, 2020. Participants 18 through 55 years of age and 56 years of age and older began enrollment from July 27, 2020, 16 through 17 years of age began enrollment from September 16, 2020, and 12 through 15 years of age began enrollment from October 15, 2020.

For participants without evidence of SARS-CoV-2 infection prior to 7 days after Dose 2, vaccine efficacy against confirmed COVID-19 occurring at least 7 days after Dose 2 was 95.0% (95% credible interval: 90.3, 97.6), which met the pre-specified success criterion. The case split was 8 COVID-19 cases in the COMIRNATY group compared to 162 COVID-19 cases in the placebo group.

The population for the updated vaccine efficacy analysis included participants 16 years of age and older who had been enrolled from July 27, 2020, and followed for the development of COVID-19 during blinded placebo-controlled follow-up through March 13, 2021, representing up to 6 months of follow-up after Dose 2. There were 12,796 (60.8%) participants in the COMIRNATY group and 12,449 (58.7%) in the placebo group followed for  $\geq 4$  months after Dose 2 in the blinded placebo-controlled follow-up period.

SARS-CoV-2 variants of concern identified from COVID-19 cases for this age group from this data cutoff include B.1.1.7 (Alpha) and B.1.351 (Beta). Representation of identified variants among cases in vaccine versus placebo recipients did not suggest decreased vaccine effectiveness against these variants.

The updated vaccine efficacy information is presented in Table 7.

**Table 7: Vaccine Efficacy – First COVID-19 Occurrence From 7 Days After Dose 2, by Age Subgroup – Participants 16 Years of Age and Older Without Evidence of Infection and Participants With or Without Evidence of Infection Prior to 7 Days After Dose 2 – Evaluable Efficacy (7 Days) Population During the Placebo-Controlled Follow-up Period**

| First COVID-19 occurrence from 7 days after Dose 2 in participants without evidence of prior SARS-CoV-2 infection* |                                                                                                                      |                                                                                                                    |                                              |
|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| Subgroup                                                                                                           | COMIRNATY<br>N <sup>a</sup> =19,993<br>Cases<br>n1 <sup>b</sup><br>Surveillance Time <sup>c</sup> (n2 <sup>d</sup> ) | Placebo<br>N <sup>a</sup> =20,118<br>Cases<br>n1 <sup>b</sup><br>Surveillance Time <sup>c</sup> (n2 <sup>d</sup> ) | Vaccine Efficacy %<br>(95% CI <sup>e</sup> ) |
| All participants                                                                                                   | 77<br>6.092 (19,711)                                                                                                 | 833<br>5.857 (19,741)                                                                                              | 91.1<br>(88.8, 93.1)                         |
| 16 through 64 years                                                                                                | 70<br>4.859 (15,519)                                                                                                 | 709<br>4.654 (15,515)                                                                                              | 90.5<br>(87.9, 92.7)                         |
| 65 years and older                                                                                                 | 7<br>1.233 (4192)                                                                                                    | 124<br>1.202 (4226)                                                                                                | 94.5<br>(88.3, 97.8)                         |



| <b>First COVID-19 occurrence from 7 days after Dose 2 in participants with or without* evidence of prior SARS-CoV-2 infection</b> |                                                                                                                           |                                                                                                                         |                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| <b>Subgroup</b>                                                                                                                   | <b>COMIRNATY<br/>N<sup>a</sup>=21,047<br/>Cases<br/>n1<sup>b</sup><br/>Surveillance Time<sup>c</sup> (n2<sup>d</sup>)</b> | <b>Placebo<br/>N<sup>a</sup>=21,210<br/>Cases<br/>n1<sup>b</sup><br/>Surveillance Time<sup>c</sup> (n2<sup>d</sup>)</b> | <b>Vaccine Efficacy %<br/>(95% CI<sup>e</sup>)</b> |
| All participants                                                                                                                  | 81<br>6.340 (20,533)                                                                                                      | 854<br>6.110 (20,595)                                                                                                   | 90.9<br>(88.5, 92.8)                               |
| 16 through 64 years                                                                                                               | 74<br>5.073 (16,218)                                                                                                      | 726<br>4.879 (16,269)                                                                                                   | 90.2<br>(87.5, 92.4)                               |
| 65 years and older                                                                                                                | 7<br>1.267 (4315)                                                                                                         | 128<br>1.232 (4326)                                                                                                     | 94.7<br>(88.7, 97.9)                               |

Note: Confirmed cases were determined by Reverse Transcription-Polymerase Chain Reaction (RT-PCR) and at least 1 symptom consistent with COVID-19 (symptoms included: fever; new or increased cough; new or increased shortness of breath; chills; new or increased muscle pain; new loss of taste or smell; sore throat; diarrhea; vomiting).

\* Participants who had no evidence of past SARS-CoV-2 infection (i.e., N-binding antibody [serum] negative at Visit 1 and SARS-CoV-2 not detected by NAAT [nasal swab] at Visits 1 and 2), and had negative NAAT (nasal swab) at any unscheduled visit prior to 7 days after Dose 2 were included in the analysis.

- N = Number of participants in the specified group.
- n1 = Number of participants meeting the endpoint definition.
- Total surveillance time in 1000 person-years for the given endpoint across all participants within each group at risk for the endpoint. Time period for COVID-19 case accrual is from 7 days after Dose 2 to the end of the surveillance period.
- n2 = Number of participants at risk for the endpoint.
- Two-sided confidence interval (CI) for vaccine efficacy is derived based on the Clopper and Pearson method adjusted to the surveillance time.

Subgroup analyses of vaccine efficacy (although limited by small numbers of cases in some subgroups) did not suggest meaningful differences in efficacy across genders, ethnic groups, geographies, or for participants with obesity or medical comorbidities associated with high risk of severe COVID-19.

### Efficacy Against Severe COVID-19

Efficacy analyses of secondary efficacy endpoints supported benefit of COMIRNATY in preventing severe COVID-19. Vaccine efficacy against severe COVID-19 is presented only for participants with or without prior SARS-CoV-2 infection (Table 8) as the COVID-19 case counts in participants without prior SARS-CoV-2 infection were the same as those in participants with or without prior SARS-CoV-2 infection in both the COMIRNATY and placebo groups.

**Table 8: Vaccine Efficacy – First Severe COVID-19 Occurrence in Participants 16 Years of Age and Older With or Without\* Prior SARS-CoV-2 Infection Based on Protocol† or Centers for Disease Control and Prevention (CDC)‡ Definition From 7 Days After Dose 2 – Evaluable Efficacy (7 Days) Population During the Placebo-Controlled Follow-up**

| <b>Vaccine Efficacy – First Severe COVID-19 Occurrence</b>                         |                                                                                                  |                                                                                                |                                                    |
|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|----------------------------------------------------|
|                                                                                    | <b>COMIRNATY<br/>Cases<br/>n1<sup>a</sup><br/>Surveillance Time<sup>b</sup> (n2<sup>c</sup>)</b> | <b>Placebo<br/>Cases<br/>n1<sup>a</sup><br/>Surveillance Time<sup>b</sup> (n2<sup>c</sup>)</b> | <b>Vaccine Efficacy %<br/>(95% CI<sup>d</sup>)</b> |
| 7 days after Dose 2 <sup>d</sup>                                                   | 1<br>6.353 (20,540)                                                                              | 21<br>6.237 (20,629)                                                                           | 95.3<br>(70.9, 99.9)                               |
| <b>Vaccine Efficacy – First Severe COVID-19 Occurrence Based on CDC Definition</b> |                                                                                                  |                                                                                                |                                                    |
|                                                                                    | <b>COMIRNATY<br/>Cases<br/>n1<sup>a</sup><br/>Surveillance Time<sup>b</sup> (n2<sup>c</sup>)</b> | <b>Placebo<br/>Cases<br/>n1<sup>a</sup><br/>Surveillance Time<sup>b</sup> (n2<sup>c</sup>)</b> | <b>Vaccine Efficacy %<br/>(95% CI<sup>d</sup>)</b> |
| 7 days after Dose 2 <sup>d</sup>                                                   | 0<br>6.345 (20,513)                                                                              | 31<br>6.225 (20,593)                                                                           | 100<br>(87.6, 100.0)                               |

Note: Confirmed cases were determined by Reverse Transcription-Polymerase Chain Reaction (RT-PCR) and at least 1 symptom consistent with COVID-19 (symptoms included: fever; new or increased cough; new or increased shortness of breath; chills; new or increased muscle pain; new loss of taste or smell; sore throat; diarrhea; vomiting).

\* Participants who had no evidence of past SARS-CoV-2 infection (i.e., N-binding antibody [serum] negative at Visit 1 and SARS-CoV-2 not detected by NAAT [nasal swab] at Visits 1 and 2), and had negative NAAT (nasal swab) at any unscheduled visit prior to 7 days after Dose 2 were included in the analysis.

† Severe illness from COVID-19 is defined in the protocol as confirmed COVID-19 and presence of at least 1 of the following:

- Clinical signs at rest indicative of severe systemic illness (respiratory rate  $\geq 30$  breaths per minute, heart rate  $\geq 125$  beats per minute, saturation of oxygen  $\leq 93\%$  on room air at sea level, or ratio of arterial oxygen partial pressure to fractional inspired oxygen  $< 300$  mm Hg);
- Respiratory failure [defined as needing high-flow oxygen, noninvasive ventilation, mechanical ventilation or extracorporeal membrane oxygenation (ECMO)];
- Evidence of shock (systolic blood pressure  $< 90$  mm Hg, diastolic blood pressure  $< 60$  mm Hg, or requiring vasopressors);
- Significant acute renal, hepatic, or neurologic dysfunction;
- Admission to an Intensive Care Unit;
- Death.

‡ Severe illness from COVID-19 as defined by CDC is confirmed COVID-19 and presence of at least 1 of the following:

- Hospitalization;
- Admission to the Intensive Care Unit;
- Intubation or mechanical ventilation;
- Death.

a. n1 = Number of participants meeting the endpoint definition.

b. Total surveillance time in 1000 person-years for the given endpoint across all participants within each group at risk for the endpoint. Time period for COVID-19 case accrual is from 7 days after Dose 2 to the end of the surveillance period.

c. n2 = Number of participants at risk for the endpoint.

d. Two-side confidence interval (CI) for vaccine efficacy is derived based on the Clopper and Pearson method adjusted to the surveillance time.

## 14.2 Efficacy in Adolescents 12 Through 15 Years of Age

A descriptive efficacy analysis of Study 2 has been performed in 2,260 adolescents 12 through 15 years of age evaluating confirmed COVID-19 cases accrued up to a data cutoff date of September 2, 2021.

The vaccine efficacy information in adolescents 12 through 15 years of age is presented in Table 9.

**Table 9: Vaccine Efficacy – First COVID-19 Occurrence From 7 Days After Dose 2: Without Evidence of Infection and With or Without Evidence of Infection Prior to 7 Days After Dose 2 – Blinded Placebo-Controlled Follow-up Period, Adolescents 12 Through 15 Years of Age Evaluable Efficacy (7 Days) Population**

| First COVID-19 occurrence from 7 days after Dose 2 in adolescents 12 through 15 years of age without evidence of prior SARS-CoV-2 infection*        |                                                                                                                    |                                                                                                                  |                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
|                                                                                                                                                     | COMIRNATY<br>N <sup>a</sup> =1057<br>Cases<br>n1 <sup>b</sup><br>Surveillance Time <sup>c</sup> (n2 <sup>d</sup> ) | Placebo<br>N <sup>a</sup> =1030<br>Cases<br>n1 <sup>b</sup><br>Surveillance Time <sup>c</sup> (n2 <sup>d</sup> ) | Vaccine Efficacy %<br>(95% CI <sup>e</sup> ) |
| Adolescents<br>12 through 15 years of age                                                                                                           | 0<br>0.343 (1043)                                                                                                  | 28<br>0.322 (1019)                                                                                               | 100.0<br>(86.8, 100.0)                       |
| First COVID-19 occurrence from 7 days after Dose 2 in adolescents 12 through 15 years of age with or without evidence of prior SARS-CoV-2 infection |                                                                                                                    |                                                                                                                  |                                              |
|                                                                                                                                                     | COMIRNATY<br>N <sup>a</sup> =1119<br>Cases<br>n1 <sup>b</sup><br>Surveillance Time <sup>c</sup> (n2 <sup>d</sup> ) | Placebo<br>N <sup>a</sup> =1109<br>Cases<br>n1 <sup>b</sup><br>Surveillance Time <sup>c</sup> (n2 <sup>d</sup> ) | Vaccine Efficacy %<br>(95% CI <sup>e</sup> ) |
| Adolescents<br>12 through 15 years of age                                                                                                           | 0<br>0.362 (1098)                                                                                                  | 30 <sup>f</sup><br>0.345 (1088)                                                                                  | 100.0<br>(87.5, 100.0)                       |

Note: Confirmed cases were determined by Reverse Transcription-Polymerase Chain Reaction (RT-PCR) and at least 1 symptom consistent with COVID-19 (symptoms included: fever; new or increased cough; new or increased shortness of breath; chills; new or increased muscle pain; new loss of taste or smell; sore throat; diarrhea; vomiting).

\* Participants who had no evidence of past SARS-CoV-2 infection (i.e., N-binding antibody [serum] negative at Visit 1 and SARS-CoV-2 not detected by NAAT [nasal swab] at Visits 1 and 2), and had negative NAAT (nasal swab) at any unscheduled visit prior to 7 days after Dose 2 were included in the analysis.

a. N = Number of participants in the specified group.

b. n1 = Number of participants meeting the endpoint definition.

c. Total surveillance time in 1000 person-years for the given endpoint across all participants within each group at risk for the endpoint. Time period for COVID-19 case accrual is from 7 days after Dose 2 to the end of the surveillance period.

d. n2 = Number of participants at risk for the endpoint.

e. Two-side confidence interval (CI) for vaccine efficacy is derived based on the Clopper and Pearson method adjusted for surveillance time.

f. The only SARS-CoV-2 variant of concern identified from COVID-19 cases in this age group from this data cutoff was B.1.1.7 (Alpha).

### 14.3 Immunogenicity in Adolescents 12 Through 15 Years of Age

In Study 2, an analysis of SARS-CoV-2 50% neutralizing titers (NT50) 1 month after Dose 2 in a randomly selected subset of participants demonstrated non-inferior immune responses (within 1.5-fold) comparing adolescents 12 through 15 years of age to participants 16 through 25 years of age who had no serological or virological evidence of past SARS-CoV-2 infection up to 1 month after Dose 2 (Table 10).

**Table 10: Summary of Geometric Mean Ratio for 50% Neutralizing Titer – Comparison of Adolescents 12 Through 15 Years of Age to Participants 16 Through 25 Years of Age (Immunogenicity Subset) – Participants Without Evidence of Infection up to 1 Month After Dose 2 – Dose 2 Evaluable Immunogenicity Population**

|                                                             |                         | COMIRNATY                                  |                                            | 12 Through 15 Years/<br>16 Through 25 Years |                                                          |
|-------------------------------------------------------------|-------------------------|--------------------------------------------|--------------------------------------------|---------------------------------------------|----------------------------------------------------------|
|                                                             |                         | 12 Through 15 Years<br>n <sup>a</sup> =190 | 16 Through 25 Years<br>n <sup>a</sup> =170 |                                             |                                                          |
| Assay                                                       | Time Point <sup>b</sup> | GMT <sup>c</sup><br>(95% CI <sup>c</sup> ) | GMT <sup>c</sup><br>(95% CI <sup>c</sup> ) | GMR <sup>d</sup><br>(95% CI <sup>d</sup> )  | Met<br>Noninferiority<br>Objective <sup>e</sup><br>(Y/N) |
| SARS-CoV-2 neutralization assay - NT50 (titer) <sup>f</sup> | 1 month after Dose 2    | 1253.6<br>(1117.7, 1406.1)                 | 708.1<br>(625.9, 801.1)                    | 1.77<br>(1.50, 2.09)                        | Y                                                        |

Abbreviations: CI = confidence interval; GMR = geometric mean ratio; GMT = geometric mean titer; LLOQ = lower limit of quantitation; NAAT = nucleic-acid amplification test; NT50 = 50% neutralizing titer; SARS-CoV-2 = severe acute respiratory syndrome coronavirus 2.

Note: Participants who had no serological or virological evidence (up to 1 month after receipt of the last dose) of past SARS-CoV-2 infection (i.e., N-binding antibody [serum] negative at Visit 1 and SARS-CoV-2 not detected by NAAT [nasal swab] at Visits 1 and 2), and had negative NAAT (nasal swab) at any unscheduled visit up to 1 month after Dose 2 were included in the analysis.

- n = Number of participants with valid and determinate assay results for the specified assay at the given dose/sampling time point.
- Protocol-specified timing for blood sample collection.
- GMTs and 2-sided 95% CIs were calculated by exponentiating the mean logarithm of the titers and the corresponding CIs (based on the Student t distribution). Assay results below the LLOQ were set to  $0.5 \times \text{LLOQ}$ .
- GMRs and 2-sided 95% CIs were calculated by exponentiating the mean difference of the logarithms of the titers (Group 1 [12 through 15 years of age] – Group 2 [16 through 25 years of age]) and the corresponding CI (based on the Student t distribution).
- Noninferiority is declared if the lower bound of the 2-sided 95% CI for the GMR is greater than 0.67.
- SARS-CoV-2 NT50 were determined using the SARS-CoV-2 mNeonGreen Virus Microneutralization Assay. The assay uses a fluorescent reporter virus derived from the USA\_WA1/2020 strain and virus neutralization is read on Vero cell monolayers. The sample NT50 is defined as the reciprocal serum dilution at which 50% of the virus is neutralized.

## 16 HOW SUPPLIED/STORAGE AND HANDLING

COMIRNATY Suspension for Intramuscular Injection, multiple dose vials with purple caps and labels with purple borders are supplied in a carton containing 25 multiple dose vials (NDC 0069-1000-03) or 195 multiple dose vials (NDC 0069-1000-02). A 0.9% Sodium Chloride Injection, USP diluent is provided but shipped separately, and should be stored at controlled room temperature 20°C to 25°C (68°F to 77°F) [see USP Controlled Room Temperature]. The provided 0.9% Sodium Chloride Injection, USP diluent will be supplied either as cartons of 10 mL single-use vials manufactured by Hospira, Inc (NDC 0409-4888-10), or 2 mL single-use vials manufactured by Fresenius Kabi USA, LLC (NDC 63323-186-02).

After dilution, 1 vial contains 6 doses of 0.3 mL.

During storage, minimize exposure to room light, and avoid exposure to direct sunlight and ultraviolet light.

Do not refreeze thawed vials.

### Frozen Vials Prior to Use

Cartons of COMIRNATY multiple dose vials with purple caps labels with purple borders arrive in thermal containers with dry ice. Once received, remove the vial cartons immediately from the thermal container and

preferably store in an ultra-low temperature freezer between -90°C to -60°C (-130°F to -76°F) until the expiry date printed on the label.

Alternatively, vials may be stored at -25°C to -15°C (-13°F to 5°F) for up to 2 weeks. Vials must be kept frozen and protected from light, in the original cartons, until ready to use. Vials stored at -25°C to -15°C (-13°F to 5°F) for up to 2 weeks may be returned 1 time to the recommended storage condition of -90°C to -60°C (-130°F to -76°F). Total cumulative time the vials are stored at -25°C to -15°C (-13°F to 5°F) should be tracked and should not exceed 2 weeks.

If an ultra-low temperature freezer is not available, the thermal container in which COMIRNATY arrives may be used as temporary storage when consistently re-filled to the top of the container with dry ice. Refer to the re-icing guidelines packed in the original thermal container for instructions regarding the use of the thermal container for temporary storage. The thermal container maintains a temperature range of -90°C to -60°C (-130°F to -76°F). Storage of the vials between -96°C to -60°C (-141°F to -76°F) is not considered an excursion from the recommended storage condition.

### Transportation of Frozen Vials

If local redistribution is needed and full cartons containing vials cannot be transported at -90°C to -60°C (-130°F to -76°F), vials may be transported at -25°C to -15°C (-13°F to 5°F). Any hours used for transport at -25°C to -15°C (-13°F to 5°F) count against the 2-week limit for storage at -25°C to -15°C (-13°F to 5°F). Frozen vials transported at -25°C to -15°C (-13°F to 5°F) may be returned 1 time to the recommended storage condition of -90°C to -60°C (-130°F to -76°F).

### Thawed Vials Before Dilution

#### *Thawed Under Refrigeration*

Thaw and then store undiluted vials in the refrigerator [2°C to 8°C (35°F to 46°F)] for up to 1 month. A carton of 25 vials or 195 vials may take up to 2 or 3 hours, respectively, to thaw in the refrigerator, whereas a fewer number of vials will thaw in less time.

#### *Thawed at Room Temperature*

For immediate use, thaw undiluted vials at room temperature [up to 25°C (77°F)] for 30 minutes. Thawed vials can be handled in room light conditions.

Vials must reach room temperature before dilution.

Undiluted vials may be stored at room temperature for no more than 2 hours.

### Transportation of Thawed Vials

Available data support transportation of 1 or more thawed vials at 2°C to 8°C (35°F to 46°F) for up to 12 hours.

### Vials After Dilution

After dilution, store vials between 2°C to 25°C (35°F to 77°F) and use within 6 hours from the time of dilution. During storage, minimize exposure to room light, and avoid exposure to direct sunlight and ultraviolet light. Any vaccine remaining in vials must be discarded after 6 hours. Do not refreeze.

## 17 PATIENT COUNSELING INFORMATION

Inform vaccine recipient of the potential benefits and risks of vaccination with COMIRNATY.

Inform vaccine recipient of the importance of completing the 2 dose vaccination series.

There is a pregnancy exposure registry for COMIRNATY. Encourage individuals exposed to COMIRNATY around the time of conception or during pregnancy to register by visiting <https://mothertobaby.org/ongoing-study/covid19-vaccines/>.

Advise vaccine recipient to report any adverse events to their healthcare provider or to the Vaccine Adverse Event Reporting System at 1-800-822-7967 and [www.vaers.hhs.gov](http://www.vaers.hhs.gov).

Prior to administering the vaccine, give the vaccine recipient the Vaccine Information Fact Sheet for Recipients and Caregivers about COMIRNATY (COVID-19 Vaccine, mRNA) and the Pfizer-BioNTech COVID-19 Vaccine to Prevent Coronavirus Disease 2019 (COVID-19) for Use in Individuals 12 Years of Age and Older. The Vaccine Information Fact Sheet for Recipients and Caregivers is available at [www.cvdvaccine-us.com](http://www.cvdvaccine-us.com).

This product's labeling may have been updated. For the most recent prescribing information, please visit <https://dailymed.nlm.nih.gov/dailymed/>.

**BIONTECH**

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